

Program : Diploma in Biomedical Engineering	
Course Code : 6249	Course Title: Biomedical Equipment Servicing and Maintenance Lab
Semester : 6	Credits: 1.5
Course Category : Program Elective	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester : 45

Course Objectives:

This course is designed to

- Provide hands-on experience in troubleshooting techniques
- Impart basic knowledge in maintenance procedures adopted for biomedical equipment.
- Inculcate the essential knowledge for testing and calibration of medical instruments.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Knowledge of handling tools		Servicing Fundamentals Lab	3
Knowledge of handling of electronic circuits		Electronic Circuits Lab	3
Knowledge of working of linear integrated circuits		Linear Integrated Circuits	4
Knowledge of working of biomedical equipment		Biomedical Instruments lab	5

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Use fundamental troubleshooting tools, procedures and circuit tracing to identify the faults in equipment.	6	Applying

CO2	Use appropriate fault diagnosis tools and techniques in analog & digital circuits.	8	Applying
CO3	Trace, draw and troubleshoot diagnostic, therapeutic equipment and clinical laboratory equipment.	20	Applying
CO4	Test and calibrate instruments using analyzers.	8	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3		3			
CO2	3	3		3			
CO3		3		3			
CO4		3		3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Use fundamental troubleshooting tools, procedures and circuit tracing to identify the faults in equipment.		
M1.01	Describe troubleshooting basics – types of faults, Troubleshooting & Maintenance aspects of the given equipment, Troubleshooting process & fault finding aids.	1	Understanding
M1.02	Test different SMD discrete components and IC packages with due care and following safety norms using proper tools/setup.	2	Applying
M1.03	Identify various input and output sockets/connectors /indicators on the given equipment.	1	Applying
M1.04	Identify various circuit boards and monitor voltages / currents at vital test points.	1	Applying
M1.05	Summarize the procedures for Test after troubleshooting, repairing and preventive maintenance steps as per NABH	1	Understanding

CO2	Use appropriate fault diagnosis tools and techniques in analog & digital equipments.		
M2.01	Apply Digital IC Troubleshooters, logic clip, logic probe, logic pulser, logic current tracer, logic comparator.	2	Applying
M2.02	Fault diagnosis in analog circuits - component testing, in-circuit testing of components, circuit tracing, signal tracing and fault identification.	2	Applying
M2.03	Tracing and Testing of voltage regulators, different types of power supplies, amplifiers etc.	2	Applying
M2.04	Identify the effects of temperature and correction with protection devices like thermister compensation, heat sink, thermal fuse, and thermal switch.	1	Applying
M2.05	Identify various connectors of biopotential recorders (common connecting probes, cables, electrodes for EEG, ECG. EMG etc), test, identify faults and repair.	1	Applying
	Lab Exam – I	1½	
CO3	Trace, draw and troubleshoot diagnostic, therapeutic equipment and clinical laboratory equipment.		
M3.01	Troubleshoot faults found in medical equipment - digital BP meter, ECG machine, Defibrillator, Pulse oximeter, Patient monitors, fetal monitor, audiometers, Infusion pump, syringe pump and nebulizer.	12	Applying
M3.02	Troubleshoot faults found in clinical laboratory equipment - pH meter, Colorimeter, Hb meter, Cell counter, auto gas analyzer etc.	8	Applying
CO4	Test and calibrate instruments using analyzers.		
M 4.01	Test and calibrate instruments using Gas flow analyzer, defibrillator analyzer, ESU analyzer and multiparameter simulator.	4	Applying
	Open Ended Experiments**	4	Applying
	Lab Exam – II	1½	

**** - Suggested Open Ended Projects**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open ended experiments as a group of 4-5. There is no duplication in experiments between groups. Open ended experiments should include the use of tools, testing and measuring instruments studied and other troubleshooting procedures.)

- Trace the circuit of faulty equipment and do the section wise analysis and find out the faults and list out the trouble shooting procedures needed.
- Test and calibrate a given instrument.
- Identify the major subsections of an instrument and assemble it and test and demonstrate the working.
- Test and rectify faults of power sockets/ power supply section/ voltage regulator section

Text / Reference:

T/R	Book Title/Author
T1	Khandpur R S, “Troubleshooting Electronic Equipment- Includes Repair & Maintenance”, Tata McGrawHill, 2nd edition, 2009.
T2	<u>A.K. Sawhney</u> <i>A Course in Electronic Measurements and Instrumentation</i> Dhanpat Rai & Co. (P) Limited 2015
R1	Dan Tomal & Neal Widmer, “Electronic Troubleshooting”, McGraw Hill, 3rd edition, 2004.
R2	Nicholas Cram & Selby Holder, “Basic Electronic Troubleshooting for Biomedical Technicians”, TSTC Publishing, 2nd edition, 2010.
R3	R S Khandpur- <i>Handbook of Biomedical instrumentation</i> - McGraw Hill Education; Third edition (4 August 2014)
R4	Ramakant A Gayakwad- <i>Op - Amps and Linear Integrated Circuits</i> Prentice Hall India Learning Private Limited 2002

Online Resources:

Sl.No	Website Link
1	https://www.vidyarthiplus.com/vp/attachment.php?aid=2447
2	https://www.electronicscircuits.com/
3	https://www.tutorialspoint.com/
4	https://www.vlab.co.in/